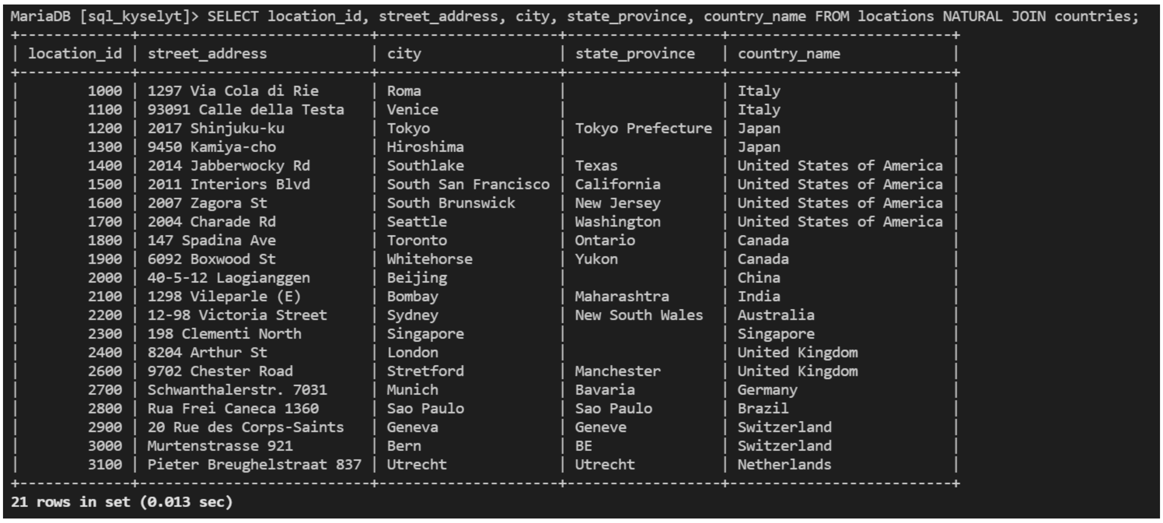
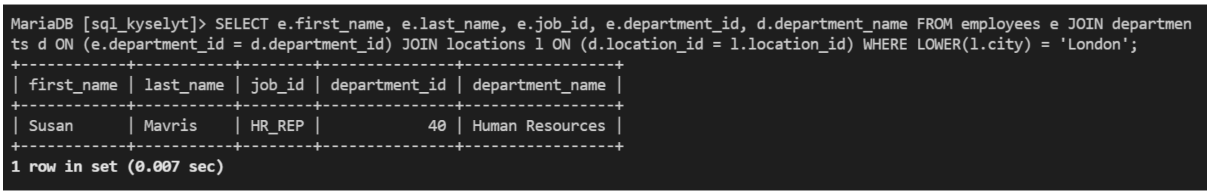
SQL-Harjoitukset (Vaativammat kyselyt)

1. Write a query to find the addresses (location_id, street_address, city, state_province, country_name) of all the departments.
   Use JOIN and tables are **departments and locations**



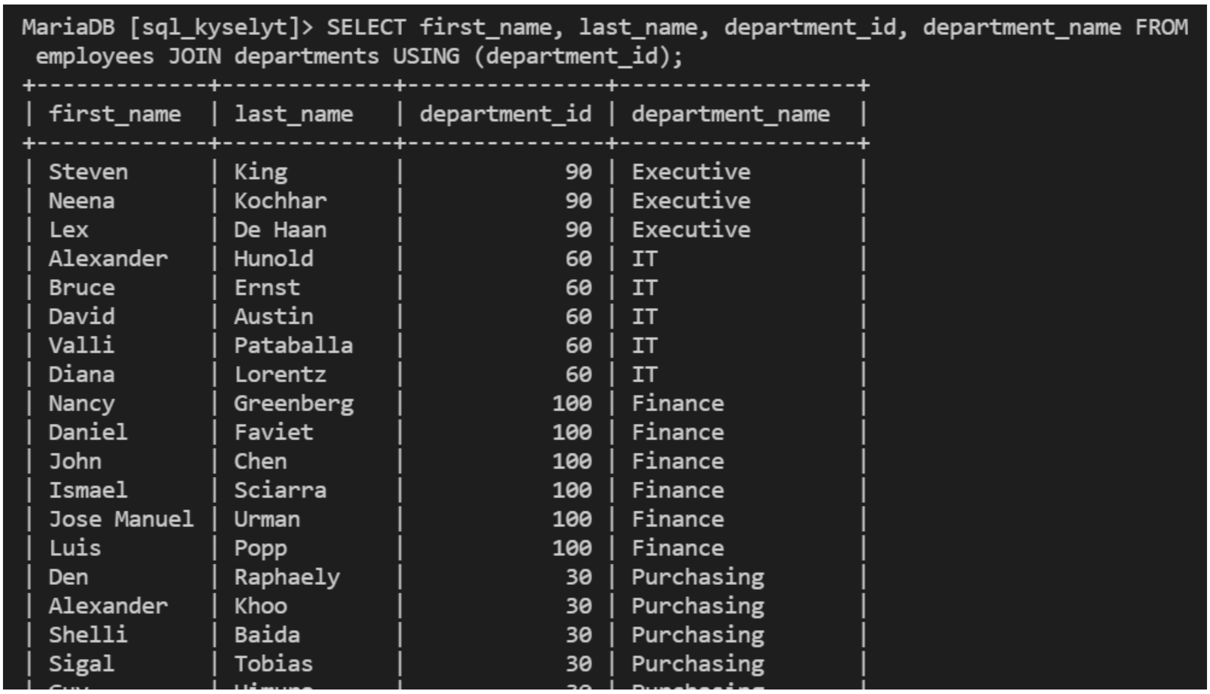
```
MariaDB [sql_kyselyt]> SELECT location_id, street_address, city, state_province, country_name FROM locations NATURAL JOIN countries;
+-------------+---------------------------+---------------------+------------------+--------------------------+
| location_id | street_address            | city                | state_province   | country_name             |
+-------------+---------------------------+---------------------+------------------+--------------------------+
|        1000 | 1297 Via Cola di Rie      | Roma                |                  | Italy                    |
|        1100 | 93091 Calle della Testa   | Venice              |                  | Italy                    |
|        1200 | 2017 Shinjuku-ku          | Tokyo               | Tokyo Prefecture | Japan                    |
|        1300 | 9450 Kamiya-cho           | Hiroshima           |                  | Japan                    |
|        1400 | 2014 Jabberwocky Rd       | Southlake           | Texas            | United States of America |
|        1500 | 2011 Interiors Blvd       | South San Francisco | California       | United States of America |
|        1600 | 2007 Zagora St            | South Brunswick     | New Jersey       | United States of America |
|        1700 | 2004 Charade Rd           | Seattle             | Washington       | United States of America |
|        1800 | 147 Spadina Ave           | Toronto             | Ontario          | Canada                   |
|        1900 | 6092 Boxwood St           | Whitehorse          | Yukon            | Canada                   |
|        2000 | 40-5-12 Laogianggen       | Beijing             |                  | China                    |
|        2100 | 1298 Vileparle (E)        | Bombay              | Maharashtra      | India                    |
|        2200 | 12-98 Victoria Street     | Sydney              | New South Wales  | Australia                |
|        2300 | 198 Clementi North        | Singapore           |                  | Singapore                |
|        2400 | 8204 Arthur St            | London              |                  | United Kingdom           |
|        2600 | 9702 Chester Road         | Stretford           | Manchester       | United Kingdom           |
|        2700 | Schwanthalerstr. 7031     | Munich              | Bavaria          | Germany                  |
|        2800 | Rua Frei Caneca 1360      | Sao Paulo           | Sao Paulo        | Brazil                   |
|        2900 | 20 Rue des Corps-Saints   | Geneva              | Geneve           | Switzerland              |
|        3000 | Murtenstrasse 921         | Bern                | BE               | Switzerland              |
|        3100 | Pieter Breughelstraat 837 | Utrecht             | Utrecht          | Netherlands              |
+-------------+---------------------------+---------------------+------------------+--------------------------+
21 rows in set (0.013 sec)
```

2. Write a query to find the names (first_name, last_name), job, department number, and department name of the employees who work in London
   Use JOIN and tables are **departments and locations**



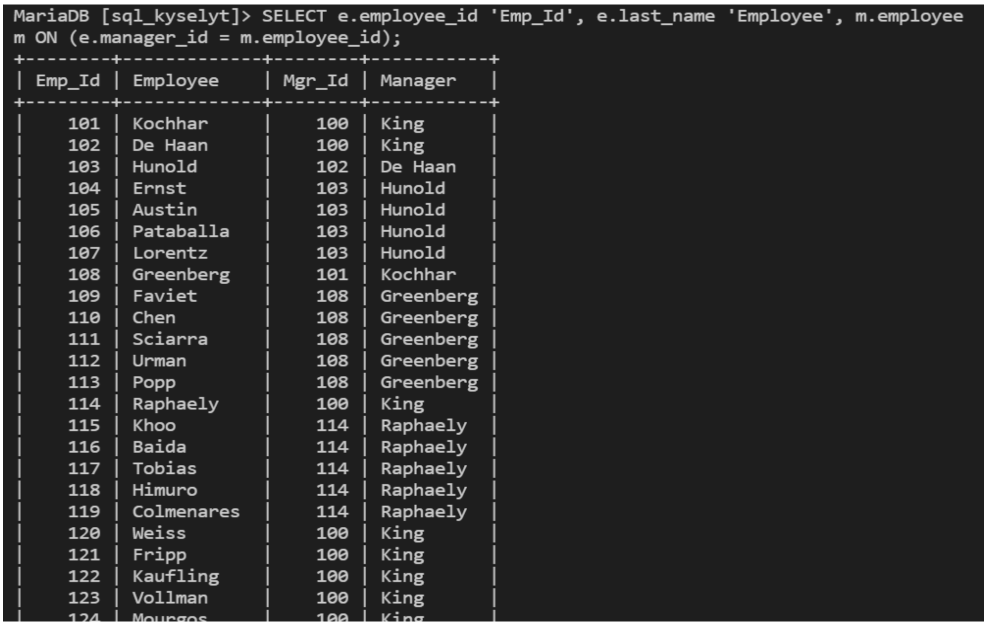
```
MariaDB [sql_kyselyt]> SELECT e.first_name, e.last_name, e.job_id, e.department_id, d.department_name FROM employees e JOIN departmen
ts d ON (e.department_id = d.department_id) JOIN locations l ON (d.location_id = l.location_id) WHERE LOWER(l.city) = 'London';
+------------+-----------+--------+---------------+-----------------+
| first_name | last_name | job_id | department_id | department_name |
+------------+-----------+--------+---------------+-----------------+
| Susan      | Mavris    | HR_REP |            40 | Human Resources |
+------------+-----------+--------+---------------+-----------------+
1 row in set (0.007 sec)
```

3. Write a query to find the names (first_name, last name), department ID and name of all the employees. Use JOIN and tables are **departments and locations**



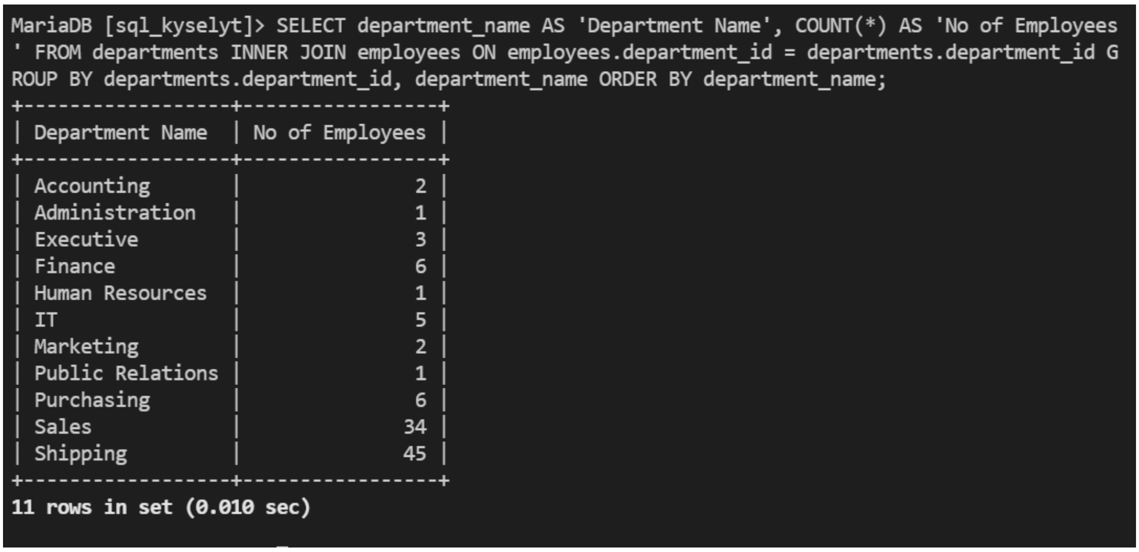
```
MariaDB [sql_kyselyt]> SELECT first_name, last_name, department_id, department_name FROM
 employees JOIN departments USING (department_id);
+-------------+-------------+---------------+------------------+
| first_name  | last_name   | department_id | department_name  |
+-------------+-------------+---------------+------------------+
| Steven      | King        |            90 | Executive        |
| Neena       | Kochhar     |            90 | Executive        |
| Lex         | De Haan     |            90 | Executive        |
| Alexander   | Hunold      |            60 | IT               |
| Bruce       | Ernst       |            60 | IT               |
| David       | Austin      |            60 | IT               |
| Valli       | Pataballa   |            60 | IT               |
| Diana       | Lorentz     |            60 | IT               |
| Nancy       | Greenberg   |           100 | Finance          |
| Daniel      | Faviet      |           100 | Finance          |
| John        | Chen        |           100 | Finance          |
| Ismael      | Sciarra     |           100 | Finance          |
| Jose Manuel | Urman       |           100 | Finance          |
| Luis        | Popp        |           100 | Finance          |
| Den         | Raphaely    |            30 | Purchasing       |
| Alexander   | Khoo        |            30 | Purchasing       |
| Shelli      | Baida       |            30 | Purchasing       |
| Sigal       | Tobias      |            30 | Purchasing       |
```

4. Write a query to find the employee id, name (last_name) along with their manager_id, manager name (last_name). Table is **employees.**



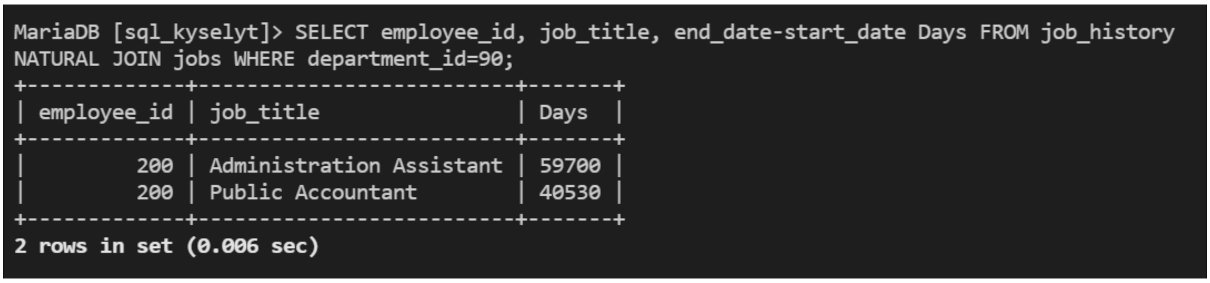
```
MariaDB [sql_kyselyt]> SELECT e.employee_id 'Emp_Id', e.last_name 'Employee', m.employee
m ON (e.manager_id = m.employee_id);
+--------+------------+--------+-----------+
| Emp_Id | Employee   | Mgr_Id | Manager   |
+--------+------------+--------+-----------+
|    101 | Kochhar    |    100 | King      |
|    102 | De Haan    |    100 | King      |
|    103 | Hunold     |    102 | De Haan   |
|    104 | Ernst      |    103 | Hunold    |
|    105 | Austin     |    103 | Hunold    |
|    106 | Pataballa  |    103 | Hunold    |
|    107 | Lorentz    |    103 | Hunold    |
|    108 | Greenberg  |    101 | Kochhar   |
|    109 | Faviet     |    108 | Greenberg |
|    110 | Chen       |    108 | Greenberg |
|    111 | Sciarra    |    108 | Greenberg |
|    112 | Urman      |    108 | Greenberg |
|    113 | Popp       |    108 | Greenberg |
|    114 | Raphaely   |    100 | King      |
|    115 | Khoo       |    114 | Raphaely  |
|    116 | Baida      |    114 | Raphaely  |
|    117 | Tobias     |    114 | Raphaely  |
|    118 | Himuro     |    114 | Raphaely  |
|    119 | Colmenares |    114 | Raphaely  |
|    120 | Weiss      |    100 | King      |
|    121 | Fripp      |    100 | King      |
|    122 | Kaufling   |    100 | King      |
|    123 | Vollman    |    100 | King      |
|    124 | Mourgos    |    100 | King      |
```

5. Write a query to get the department name and number of employees in the department and order result by department name. Table are **employees and departments. Use INNER JOIN**



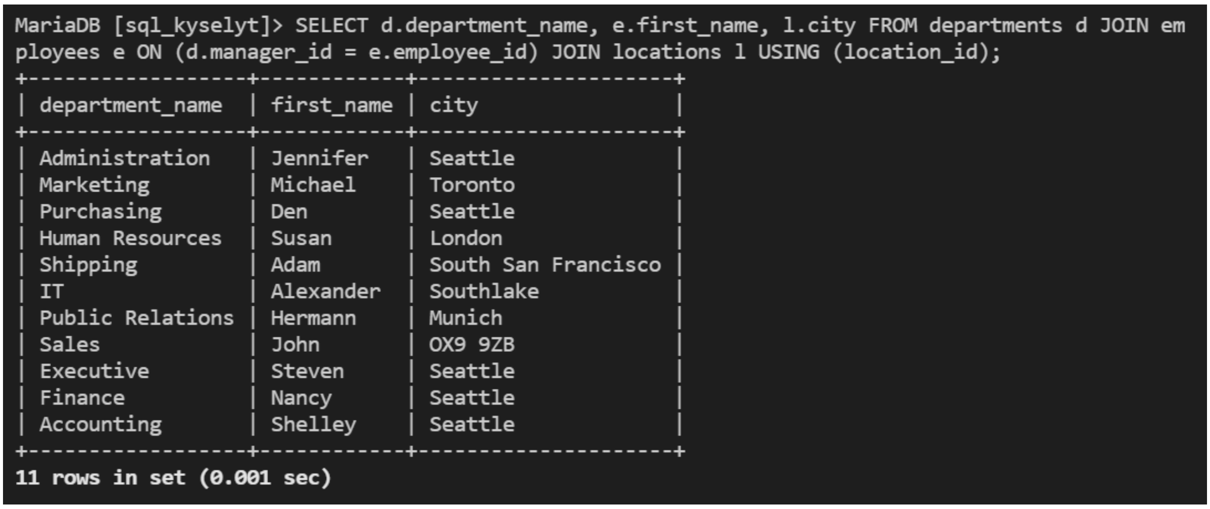
```
MariaDB [sql_kyselyt]> SELECT department_name AS 'Department Name', COUNT(*) AS 'No of Employees
' FROM departments INNER JOIN employees ON employees.department_id = departments.department_id G
ROUP BY departments.department_id, department_name ORDER BY department_name;
+------------------+-----------------+
| Department Name  | No of Employees |
+------------------+-----------------+
| Accounting       |               2 |
| Administration   |               1 |
| Executive        |               3 |
| Finance          |               6 |
| Human Resources  |               1 |
| IT               |               5 |
| Marketing        |               2 |
| Public Relations |               1 |
| Purchasing       |               6 |
| Sales            |              34 |
| Shipping         |              45 |
+------------------+-----------------+
11 rows in set (0.010 sec)
```

6. Find the employee ID, job title, number of days between ending date and starting date for all jobs in department 90 from job history. Table is **employees. Use NATURAL JOIN AND WHERE depatment_id=90**



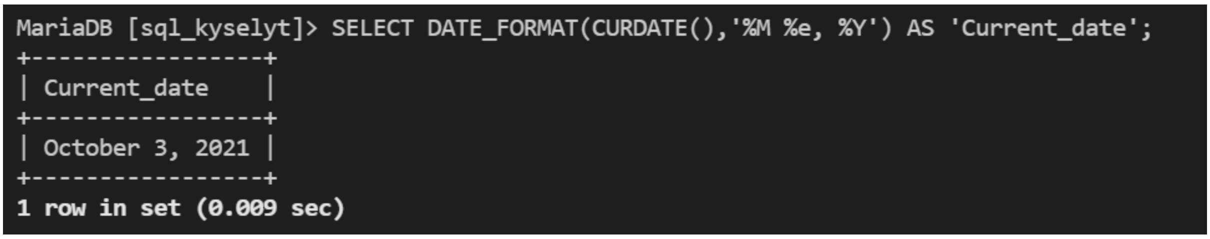
```
MariaDB [sql_kyselyt]> SELECT employee_id, job_title, end_date-start_date Days FROM job_history
NATURAL JOIN jobs WHERE department_id=90;
+-------------+-------------------------+-------+
| employee_id | job_title               | Days  |
+-------------+-------------------------+-------+
|         200 | Administration Assistant | 59700 |
|         200 | Public Accountant       | 40530 |
+-------------+-------------------------+-------+
2 rows in set (0.006 sec)
```

7. Write a query to display the department name, manager name, and city. Tables are departments, locations and employees.



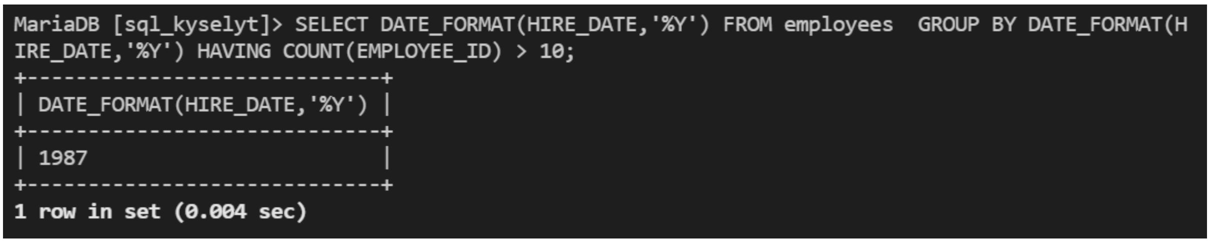
```
MariaDB [sql_kyselyt]> SELECT d.department_name, e.first_name, l.city FROM departments d JOIN em
ployees e ON (d.manager_id = e.employee_id) JOIN locations l USING (location_id);
+------------------+------------+---------------------+
| department_name  | first_name | city                |
+------------------+------------+---------------------+
| Administration   | Jennifer   | Seattle             |
| Marketing        | Michael    | Toronto             |
| Purchasing       | Den        | Seattle             |
| Human Resources  | Susan      | London              |
| Shipping         | Adam       | South San Francisco |
| IT               | Alexander  | Southlake           |
| Public Relations | Hermann    | Munich              |
| Sales            | John       | OX9 9ZB             |
| Executive        | Steven     | Seattle             |
| Finance          | Nancy      | Seattle             |
| Accounting       | Shelley    | Seattle             |
+------------------+------------+---------------------+
11 rows in set (0.001 sec)
```

8. Write a query to get the current date in the following format.



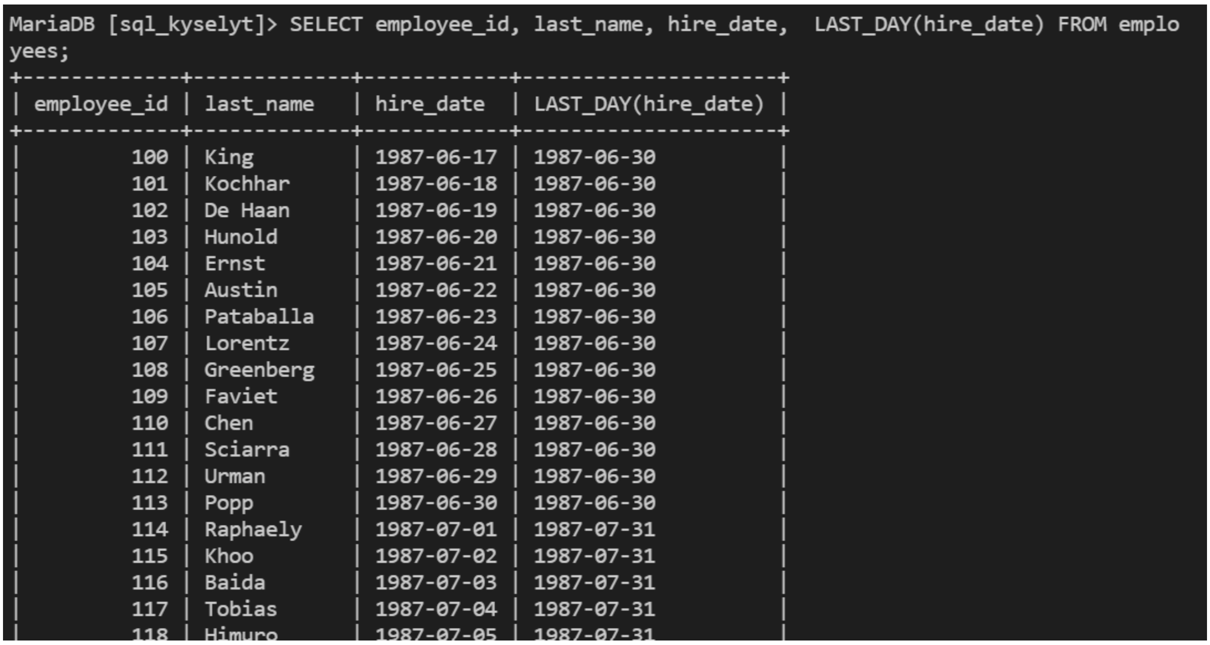
```
MariaDB [sql_kyselyt]> SELECT DATE_FORMAT(CURDATE(),'%M %e, %Y') AS 'Current_date';
+-----------------+
| Current_date    |
+-----------------+
| October 3, 2021 |
+-----------------+
1 row in set (0.009 sec)
```

9. Write a query to get the years in which more than 10 employees joined. Table is job_history.



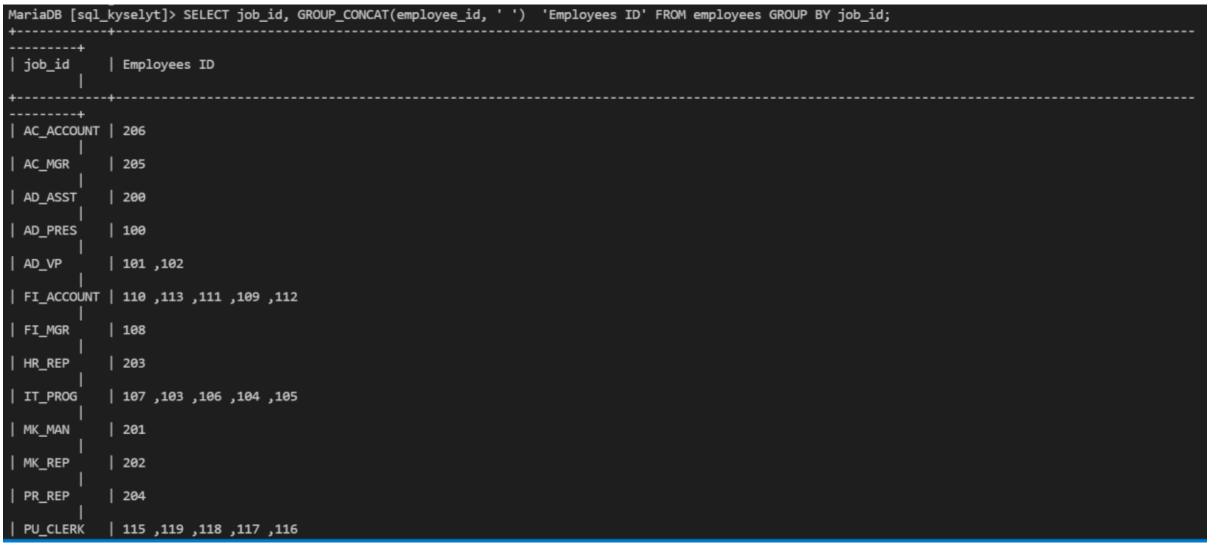
```
MariaDB [sql_kyselyt]> SELECT DATE_FORMAT(HIRE_DATE,'%Y') FROM employees  GROUP BY DATE_FORMAT(H
IRE_DATE,'%Y') HAVING COUNT(EMPLOYEE_ID) > 10;
+-----------------------------+
| DATE_FORMAT(HIRE_DATE,'%Y') |
+-----------------------------+
| 1987                        |
+-----------------------------+
1 row in set (0.004 sec)
```

10. Write a query to get employee ID, last name, and date of first salary of the employees. Table is employees.



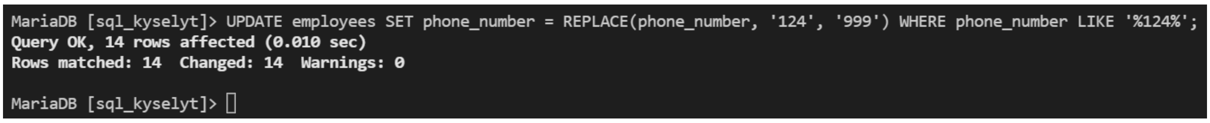
```
MariaDB [sql_kyselyt]> SELECT employee_id, last_name, hire_date,  LAST_DAY(hire_date) FROM emplo
yees;
+-------------+-------------+------------+---------------------+
| employee_id | last_name   | hire_date  | LAST_DAY(hire_date) |
+-------------+-------------+------------+---------------------+
|         100 | King        | 1987-06-17 | 1987-06-30          |
|         101 | Kochhar     | 1987-06-18 | 1987-06-30          |
|         102 | De Haan     | 1987-06-19 | 1987-06-30          |
|         103 | Hunold      | 1987-06-20 | 1987-06-30          |
|         104 | Ernst       | 1987-06-21 | 1987-06-30          |
|         105 | Austin      | 1987-06-22 | 1987-06-30          |
|         106 | Pataballa   | 1987-06-23 | 1987-06-30          |
|         107 | Lorentz     | 1987-06-24 | 1987-06-30          |
|         108 | Greenberg   | 1987-06-25 | 1987-06-30          |
|         109 | Faviet      | 1987-06-26 | 1987-06-30          |
|         110 | Chen        | 1987-06-27 | 1987-06-30          |
|         111 | Sciarra     | 1987-06-28 | 1987-06-30          |
|         112 | Urman       | 1987-06-29 | 1987-06-30          |
|         113 | Popp        | 1987-06-30 | 1987-06-30          |
|         114 | Raphaely    | 1987-07-01 | 1987-07-31          |
|         115 | Khoo        | 1987-07-02 | 1987-07-31          |
|         116 | Baida       | 1987-07-03 | 1987-07-31          |
|         117 | Tobias      | 1987-07-04 | 1987-07-31          |
|         118 | Himuro      | 1987-07-05 | 1987-07-31          |
```

11. Write a query to get the job_id and related employee's id. Table is employees. Use GROUP_CONCAT command.



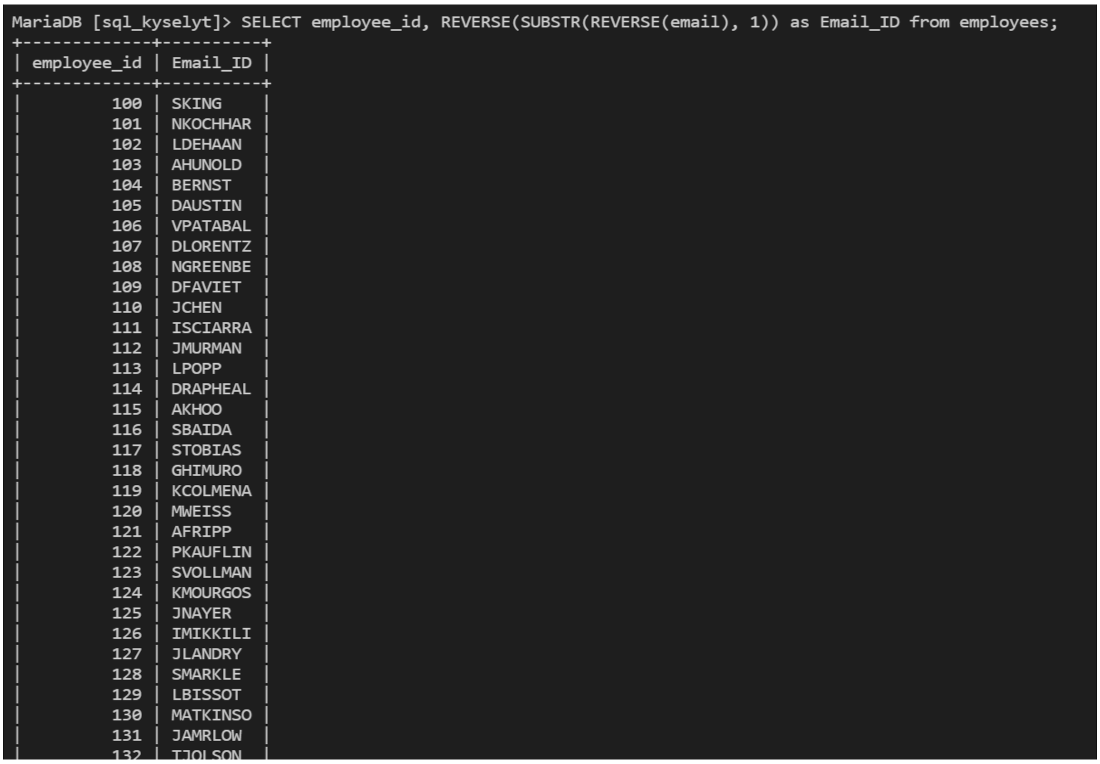
```
MariaDB [sql_kyselyt]> SELECT job_id, GROUP_CONCAT(employee_id, ' ')  'Employees ID' FROM employees GROUP BY job_id;
+------------+------------------------------------------------------------------------------------------------------------------------------------------------
---------+
| job_id     | Employees ID
         |
+------------+------------------------------------------------------------------------------------------------------------------------------------------------
---------+
| AC_ACCOUNT | 206
         |
| AC_MGR     | 205
         |
| AD_ASST    | 200
         |
| AD_PRES    | 100
         |
| AD_VP      | 101 ,102
         |
| FI_ACCOUNT | 110 ,113 ,111 ,109 ,112
         |
| FI_MGR     | 108
         |
| HR_REP     | 203
         |
| IT_PROG    | 107 ,103 ,106 ,104 ,105
         |
| MK_MAN     | 201
         |
| MK_REP     | 202
         |
| PR_REP     | 204
         |
| PU_CLERK   | 115 ,119 ,118 ,117 ,116
```

12. Write a query to update the portion of the phone_number in the employees table, within the phone number the substring '124' will be replaced by '999'. Use REPLACE and LIKE %124% to find current string.



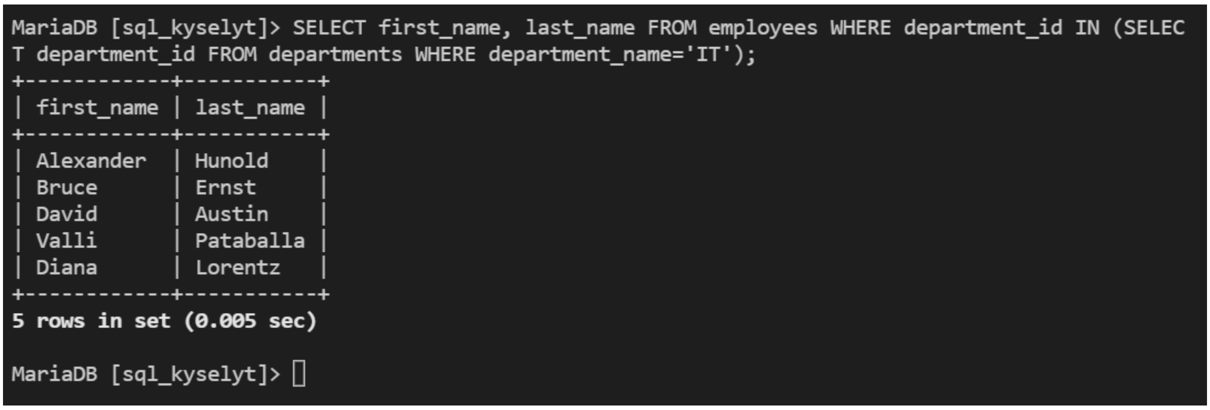
```
MariaDB [sql_kyselyt]> UPDATE employees SET phone_number = REPLACE(phone_number, '124', '999') WHERE phone_number LIKE '%124%';
Query OK, 14 rows affected (0.010 sec)
Rows matched: 14  Changed: 14  Warnings: 0

MariaDB [sql_kyselyt]> 
```

13. Write a query to get the employee id, email id (discard 0 characters).



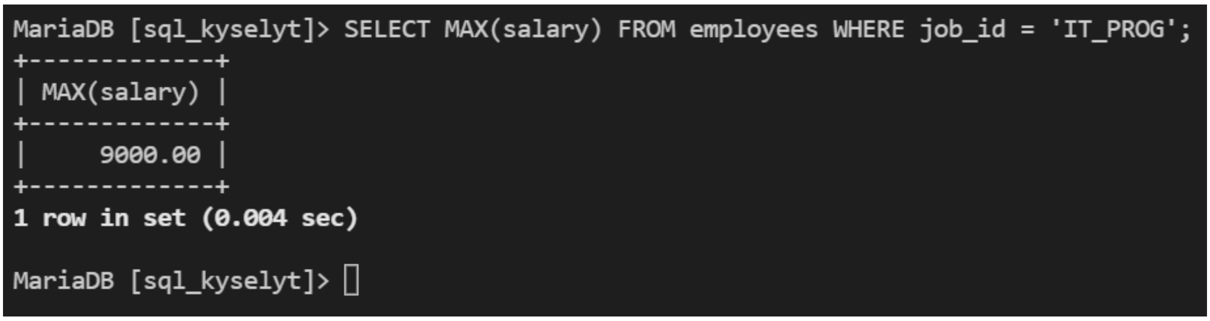
```
MariaDB [sql_kyselyt]> SELECT employee_id, REVERSE(SUBSTR(REVERSE(email), 1)) as Email_ID from employees;
+-------------+----------+
| employee_id | Email_ID |
+-------------+----------+
|         100 | SKING    |
|         101 | NKOCHHAR |
|         102 | LDEHAAN  |
|         103 | AHUNOLD  |
|         104 | BERNST   |
|         105 | DAUSTIN  |
|         106 | VPATABAL |
|         107 | DLORENTZ |
|         108 | NGREENBE |
|         109 | DFAVIET  |
|         110 | JCHEN    |
|         111 | ISCIARRA |
|         112 | JMURMAN  |
|         113 | LPOPP    |
|         114 | DRAPHEAL |
|         115 | AKHOO    |
|         116 | SBAIDA   |
|         117 | STOBIAS  |
|         118 | GHIMURO  |
|         119 | KCOLMENA |
|         120 | MWEISS   |
|         121 | AFRIPP   |
|         122 | PKAUFLIN |
|         123 | SVOLLMAN |
|         124 | KMOURGOS |
|         125 | JNAYER   |
|         126 | IMIKKILI |
|         127 | JLANDRY  |
|         128 | SMARKLE  |
|         129 | LBISSOT  |
|         130 | MATKINSO |
|         131 | JAMRLOW  |
|         132 | TJOLSON  |
```

14. Find the names (first_name, last_name) of all employees who works in the IT department. Use subquery



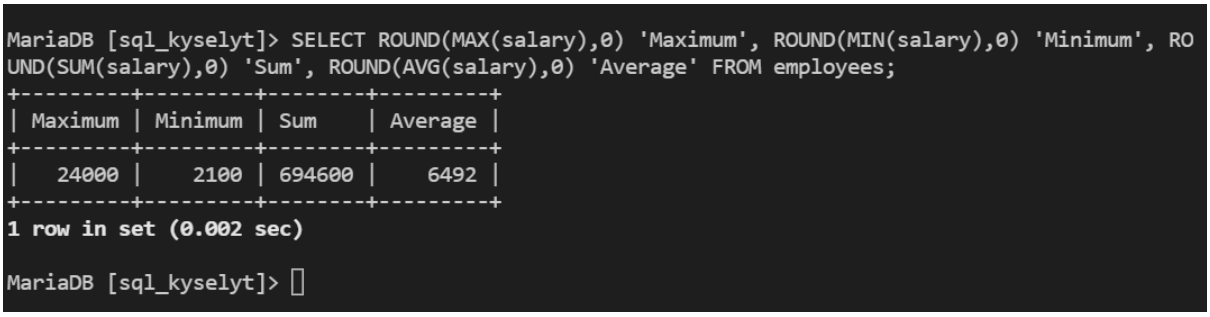
```
MariaDB [sql_kyselyt]> SELECT first_name, last_name FROM employees WHERE department_id IN (SELEC
T department_id FROM departments WHERE department_name='IT');
+------------+-----------+
| first_name | last_name |
+------------+-----------+
| Alexander  | Hunold    |
| Bruce      | Ernst     |
| David      | Austin    |
| Valli      | Pataballa |
| Diana      | Lorentz   |
+------------+-----------+
5 rows in set (0.005 sec)

MariaDB [sql_kyselyt]>
```

15. Write a query to get the maximum salary of an employee working as a Programmer. Job_id is IT_PROG.



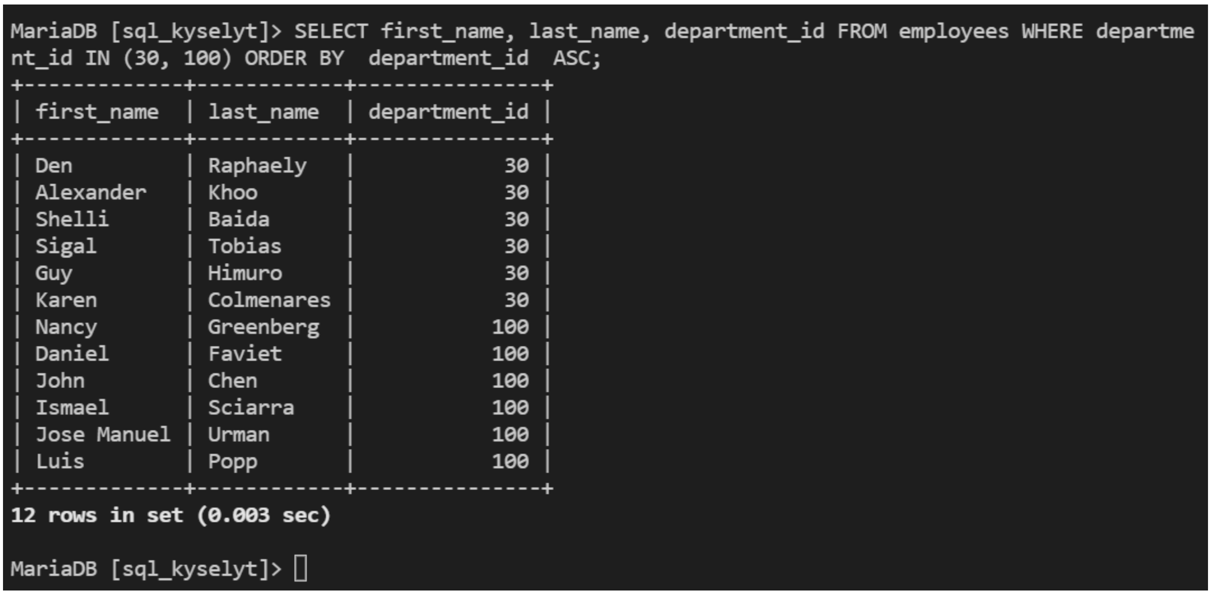
```
MariaDB [sql_kyselyt]> SELECT MAX(salary) FROM employees WHERE job_id = 'IT_PROG';
+-------------+
| MAX(salary) |
+-------------+
|     9000.00 |
+-------------+
1 row in set (0.004 sec)

MariaDB [sql_kyselyt]>
```

16. Write a query to get the highest, lowest, sum, and average salary of all employees.



```
MariaDB [sql_kyselyt]> SELECT ROUND(MAX(salary),0) 'Maximum', ROUND(MIN(salary),0) 'Minimum', RO
UND(SUM(salary),0) 'Sum', ROUND(AVG(salary),0) 'Average' FROM employees;
+---------+---------+--------+---------+
| Maximum | Minimum | Sum    | Average |
+---------+---------+--------+---------+
|   24000 |    2100 | 694600 |    6492 |
+---------+---------+--------+---------+
1 row in set (0.002 sec)

MariaDB [sql_kyselyt]>
```

17. Write a query to display the names (first_name, last_name) and department ID of all employees in departments 30 or 100 in ascending alphabetical order by department ID.



```
MariaDB [sql_kyselyt]> SELECT first_name, last_name, department_id FROM employees WHERE departme
nt_id IN (30, 100) ORDER BY  department_id  ASC;
+-------------+------------+---------------+
| first_name  | last_name  | department_id |
+-------------+------------+---------------+
| Den         | Raphaely   |            30 |
| Alexander   | Khoo       |            30 |
| Shelli      | Baida      |            30 |
| Sigal       | Tobias     |            30 |
| Guy         | Himuro     |            30 |
| Karen       | Colmenares |            30 |
| Nancy       | Greenberg  |           100 |
| Daniel      | Faviet     |           100 |
| John        | Chen       |           100 |
| Ismael      | Sciarra    |           100 |
| Jose Manuel | Urman      |           100 |
| Luis        | Popp       |           100 |
+-------------+------------+---------------+
12 rows in set (0.003 sec)

MariaDB [sql_kyselyt]>
```